part of TUM THINK TANK

QSL

RESPONSIBILITY IN DESIGN CHOICES

What the model decides for others



Presented by **Gabriela Boechat**

https://tumthinktank.de/project/quantum-social-lab/

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FRAMING

Every model decides something for someone else.

Most of these decisions are made during design, long before anyone is affected. How we design and develop a system can shape its consequences for society (Kop et al., 2024).

Today we ask three questions about each decision:

Who is affected?

Who might be left out?

Who would know if something went wrong?



FOUR DECISIONS

GovTech

A state opens a regulatory sandbox - a testing space with relaxed rules - for quantum technologies. Companies are admitted by a readiness score based on technical benchmarks. Small teams that use cloud access instead of their own hardware score lower.

Life Sciences

A quantum sensor classifies readings as normal or abnormal based on a cutoff. The cutoff was set with data from one hospital, and sensor noise led the team to set it conservatively. A regional hospital with older patients now wants to use it.

Mobility

A quantum-assisted forecast determines which bus lines receive more or fewer buses. Current hardware can only handle a simplified model, so the city is divided into 12 zones and each zone gets one average. Lines with low predicted demand lose buses.

Finance

A quantum risk model places customers into risk bands that affect their access to credit. The model learned from past lending decisions, and its results are estimates that a regulator cannot easily reproduce. Customers in the lowest band are refused or pay more.

GROUP DISCUSSION

What does the model decide?

What is decided on someone else's behalf and who lives with the result?

Who might be left out?

Who could be classified differently or receive less than they need?

What does the model not see?

What does it ignore or simplify and what was given up to make it work?

Who would notice if something went wrong?

Would there be a record? Who would see it?

What would you tell the person who has to act?

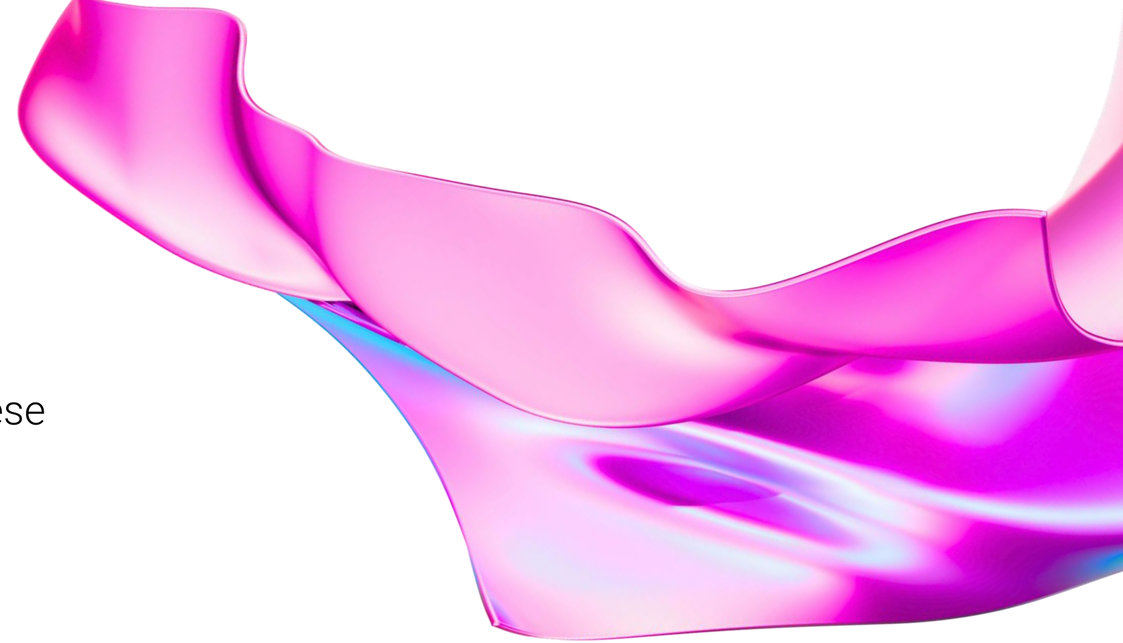
What does the single number hide, and how would you say it in 30 seconds?

15 minutes discussion, 5 minutes per group for presentations



RESPONSIBLE QUANTUM INNOVATION STARTS DURING DEVELOPMENT

Kop et al. propose ten principles for responsible quantum innovation. These three connect technical choices to societal and institutional decisions.



Inclusion

Whose knowledge and experience shape priorities, design and testing?

Diverse disciplinary, professional and social perspectives can expose concerns while design choices remain revisable.

Societal Relevance

Which societal goal does this serve, who benefits, and could it deepen inequality?

Direct quantum R&D towards desirable societal goals and assess whether its benefits and harms are distributed fairly.

Responsibility

How will the ecosystem learn from uses and consequences?

Build feedback loops that track risks and opportunities at the application level, support learning from use and avoid concentrating the costs of that learning on vulnerable groups.

Take one question for the Sprint phase: what should your design take into account, beyond the technical challenge?

Source: Kop et al. (2024), Principles 6, 7 and 9, pp. 6–9

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THANK YOU



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